

Let the Sheet Do the Work — Paper B

MEDIUM • NO COMPUTER • NO CALCULATOR • 35 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. There is no computer and no calculator: every number is designed to work by hand.
- This paper is about **Google Sheets**. Every formula starts with **=**, arguments are separated by **commas**, and text goes in **quotation marks**.
- Where a question asks for a **value**, write what the cell would show. Where it asks for a **formula**, write it exactly, brackets and commas included.
- The mark for each question is shown in square brackets on the right.
- **Reminder (medium papers only).** SUM(range) AVERAGE(range) MIN(range) MAX(range) COUNT(range) COUNTA(range) MINUS(a, b) DIVIDE(a, b) PRODUCT(a, b) IF(test, if_true, if_false) SUMIF(range, criterion, sum_range) COUNTIF(range, criterion)

SECTION 1 – DATA, STEPS AND FUNCTIONS

1. The six steps of data analytics are listed below out of order. In the right-hand column, write 1 to 6 to show the order in which they happen. [3]

	Step of data analytics	Write 1 to 6
	Process — run formulas for totals, averages, counts	
	Present — a chart or a sentence someone can act on	
	Collect — gather till slips, a survey, sensor readings	
	Organise — one row per thing, one column per fact, a header row	
	Analyse — compare groups, find the odd one out	
	Clean — fix the typo, the blank, the wrong spelling	

2. A canteen sheet has one row that reads **Masala Chai, 150**.

- (a) Which of these is **data** and which is **information**? (i) “Masala Chai, 150” (ii) “Masala Chai sold more units than any other item.” [1]

- (b) State what has to be done to data to turn it into information. [1]

3. Rainfall in millimetres is held in this sheet. Write the value each formula would show.

	A	B	C	D
1	City	Jun	Jul	Aug
2	Mumbai	540	840	590
3	Pune	130	180	150
4	Chennai	60	90	120

(a) =SUM(B2:D2) [1]

(b) =MAX(B2:B4) [1]

(c) =AVERAGE(C2:C4) [1]

(d) =MINUS(D2,D4) [1]

(e) =MIN(B2:D4) [1]

SECTION 2 – CONDITIONS

4. The canteen sheet below has the Category in column B and the units sold in column C, rows 2 to 8.

	A	B	C
1	Item	Category	Units
2	Samosa	Snack	24
3	Idli	Meal	17
4	Chai	Drink	30
5	Poha	Meal	12
6	Lassi	Drink	18
7	Puff	Snack	16
8	Biryani	Meal	9

(a) Write a formula that counts the items in the **Meal** category, and state its value. [2]

- (b) Write a formula that adds up the units sold in the **Snack** category, and state its value. [2]
-
-

5. Marks out of 40 are in column B. Column C should show **Gold** for 34 or more, **Silver** for 26 to 33, **Bronze** for 16 to 25, and **None** below 16. Fill the three gaps in the formula for C2. [3]

=IF(B2>=34,"Gold",IF(B2>=____,"Silver",IF(B2>=____,"Bronze","____")))

6. Another tutor writes this formula in C2 for marks out of 50, intending 42+ Distinction, 35 to 41 Merit, 20 to 34 Pass, and below 20 Fail:

=IF(B2>=20,"Pass",IF(B2>=35,"Merit",IF(B2>=42,"Distinction","Fail")))

- (a) Write what the formula shows for each mark. [2]

	A	B
1	Mark in B2	The formula shows
2	47	
3	36	
4	25	
5	12	

- (b) Explain what is wrong with the formula. [1]
-
-

- 7.** A Subject column was typed by hand. The extract is below; the space typed after the word in row 6 is shown as ␣.

	A	B	C
1	Student	Subject	Marks
2	Anaya	Science	33
3	Bilal	SCIENCE	41
4	Chitra	Sci	28
5	Deven	English	36
6	Esha	Science␣	45
7	Farid	Science	38
8	Gita	English	30

- (a)** State the value of =COUNTIF(B2:B8, "Science"). [1]

- (b)** Rows 4 and 6 are both meant to be Science. Say why COUNTIF does not count either of them. [1]

- (c)** Design a data validation rule for column B that would have stopped both. State the criteria and the setting for invalid data. [1]

- (d)** The rule is now added. State what the formula in (a) shows, and what must be done to make it show 5. [1]

8. Six branches of a shop record their takings in rupees. Zone is in column B and Takings in column C, rows 2 to 7.

	A	B	C
1	Branch	Zone	Takings (₹)
2	Hebbal	North	400
3	Jayanagar	South	300
4	Yelahanka	North	500
5	Banashankari	South	200
6	Koramangala	South	250
7	Malleshwaram	North	450

- (a) Write a formula that adds up the takings of the North zone. [1]

- (b) Using SUMIF and COUNTIF, write one formula for the **average** takings of a North branch, and state its value. [2]

9. Choose a chart type for each, and give a reason.

- (a) The total takings of the North zone and of the South zone, side by side. [1]

- (b) Mumbai's rainfall in June, July, August and September. [1]

- (c) A chart has been drawn with neither a title nor axis labels. Say what must be added. [1]

End of paper. Check that every formula starts with = and that every word inside a formula is in quotation marks.